

EVALUATING ENM PERFORMANCE WITH NULL MODELS

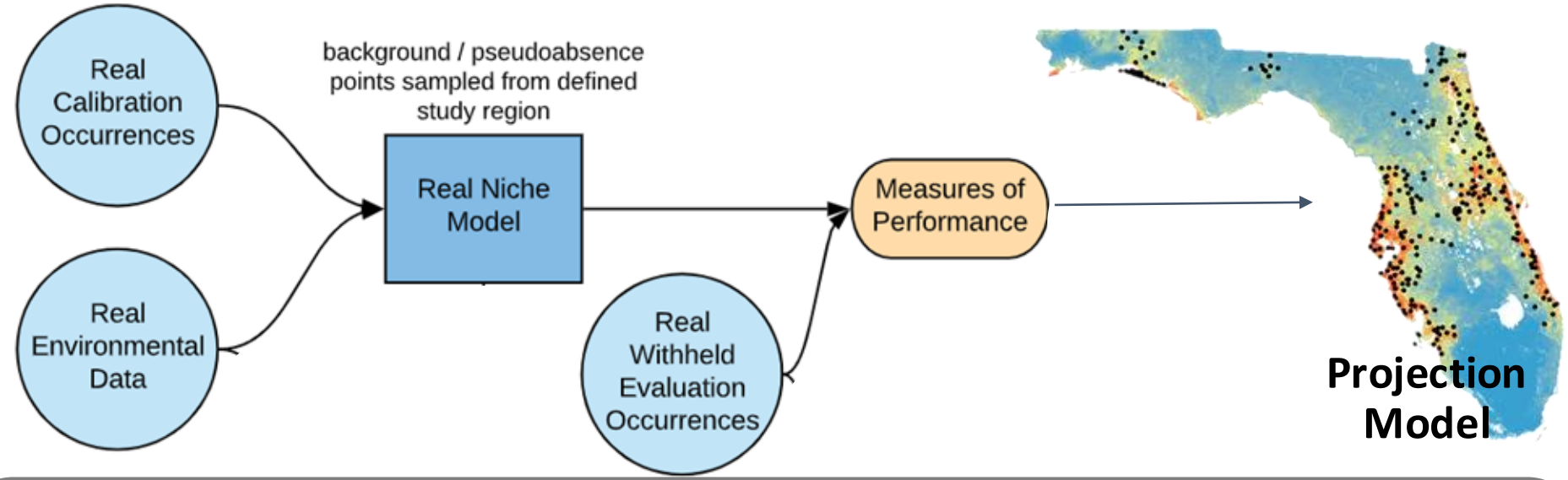
Slides by Tyler Radtke
Edited & Presented by JT Miller
University of Florida

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iDigBio
Integrated Digitized Biocollections

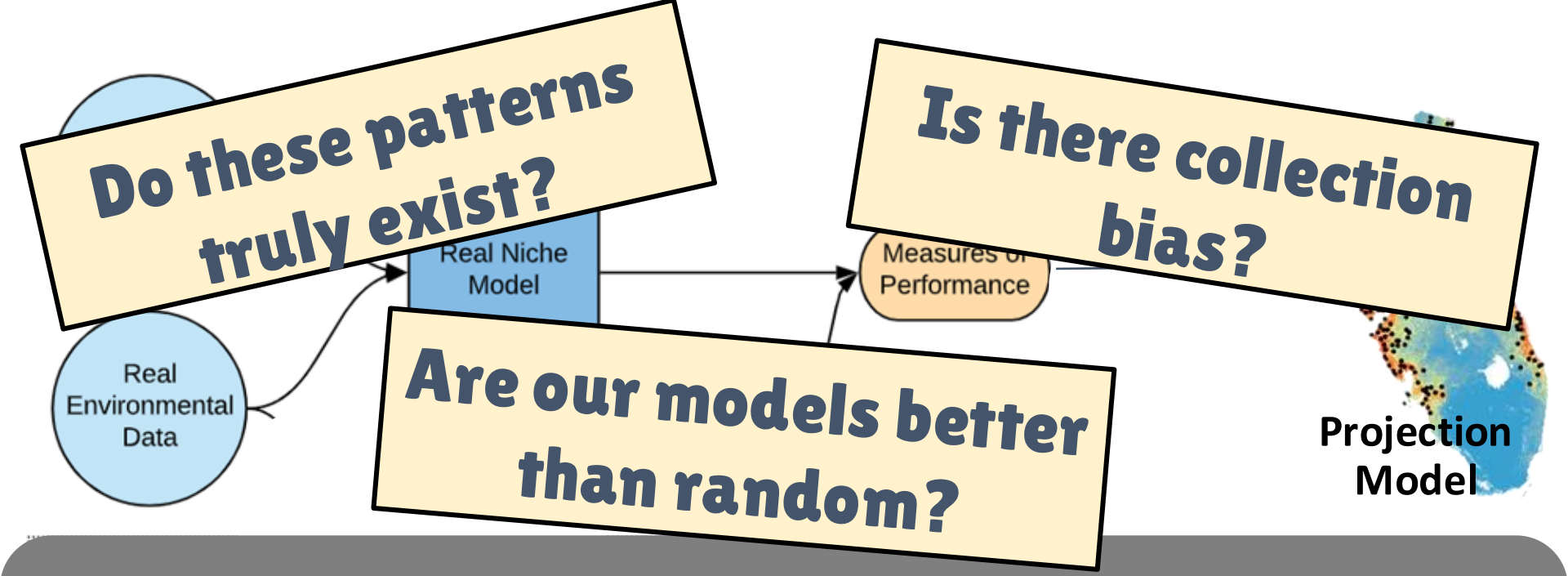
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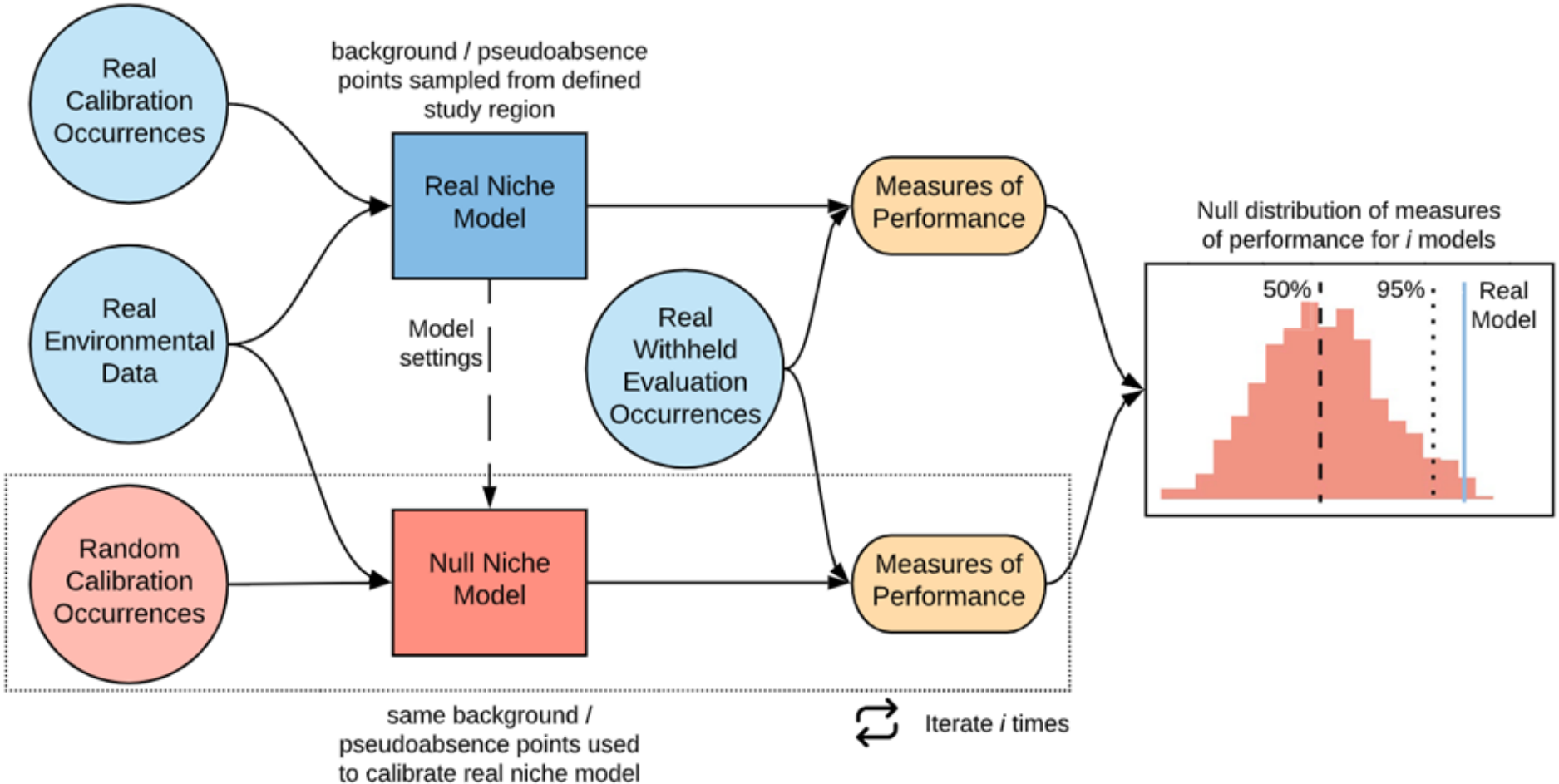
The Typical ENM Process

Bohl, C. L., Kass, J. M., & Anderson, R. P. (2019). A new null model approach to quantify performance and significance for ecological niche models of species distributions. *Journal of Biogeography*, *46*(6), 1101–1111. https://doi.org/10.1111/jbi.13573



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Actual vs. predicted habitat suitability values

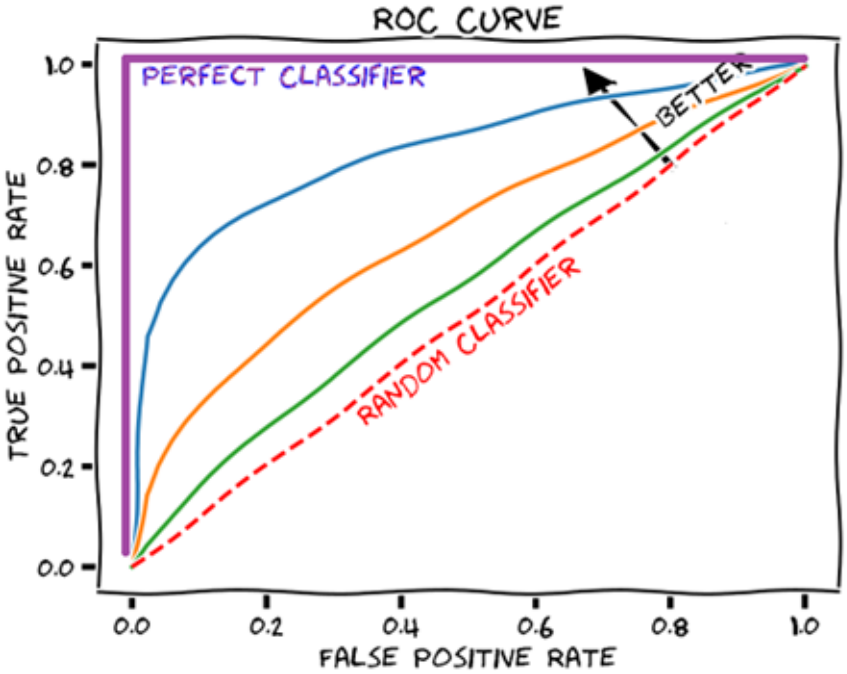
PREDICTED VALUES

ACTUAL VALUES	Positive	Negative
Positive	True Positive (TP)	False Negative (FN)
Negative	False Positive (FP)	True Negative (TN)

PREDICTED VALUES

ACTUAL VALUES	Apple	Strawberry
Apple		
Strawberry		

Area Under the Curve (AUC)



False positive rate (1 - specificity) vs. true positive rate (sensitivity)

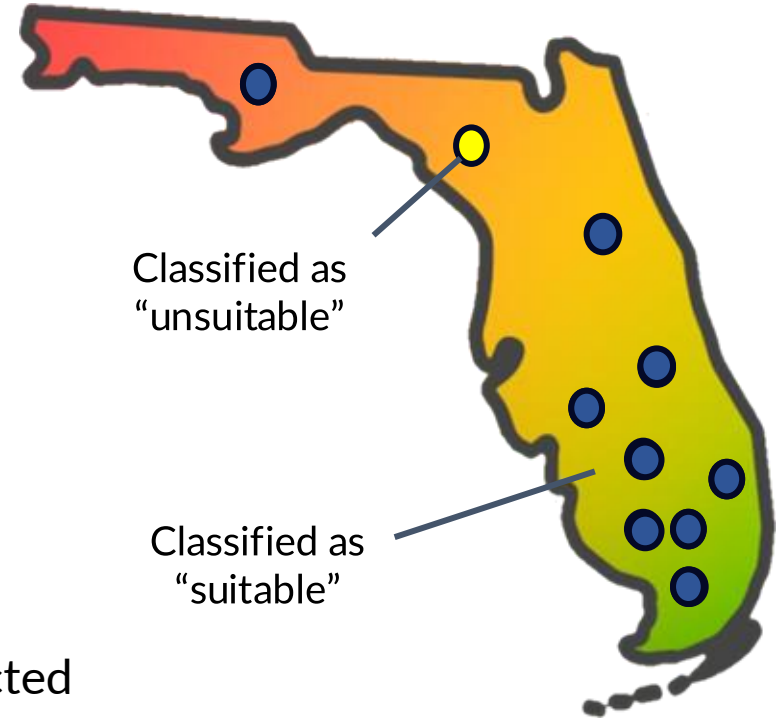
AUC = 1.0 -> perfect model
AUC = 0.5 -> model performs no better than random
AUC = <0.5 -> a sign of error

AUC evaluates a model's ability to distinguish between presence/background points

Omission Rate (OR)

OR = the proportion of test/validation presence points that fall below a specific threshold in training.

Example Threshold:
OR = 0.1
OR > 0.1 Possible Overfitting
OR < 0.1 Doing better than expected

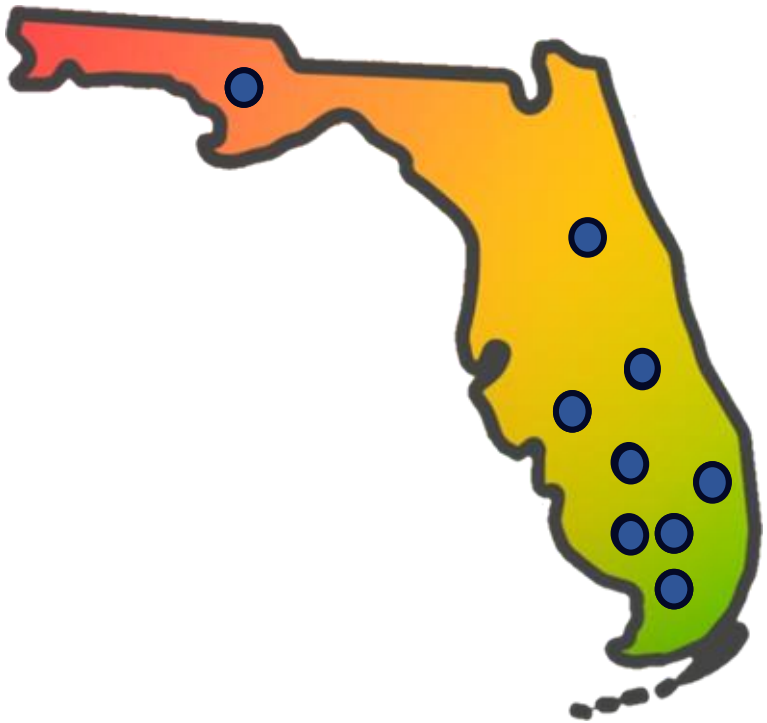


Continuous Boyce Index (CBI)

Presence-only performance metric based on the relationship between predicted suitability and where presences actually occur.

In ENMeval, its calculated by comparing the distribution of suitability at presences vs background across the continuous suitability range, and is summarized in a Spearman Correlation

CBI: -1:1



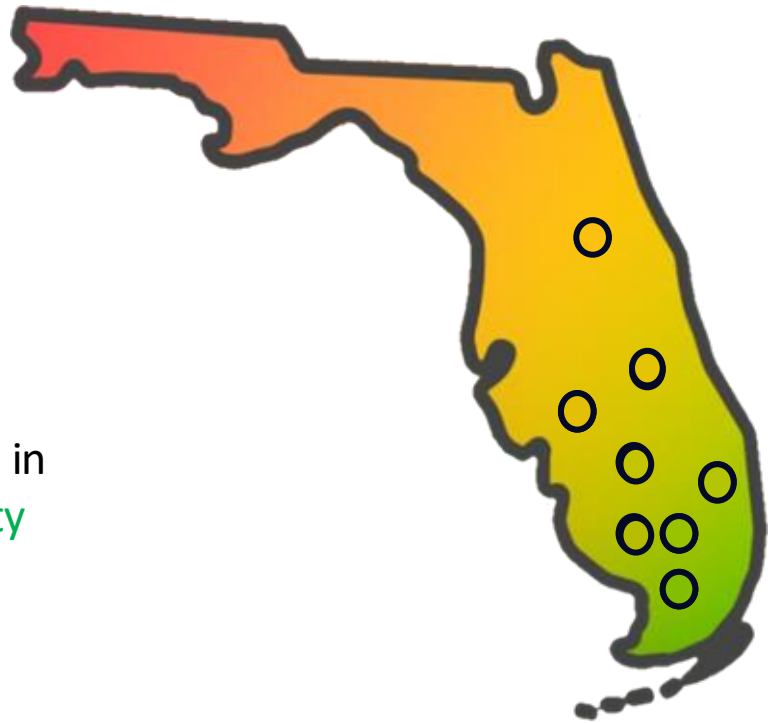
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$$CBI \approx 1$$

Presences concentrated in high suitability habitats



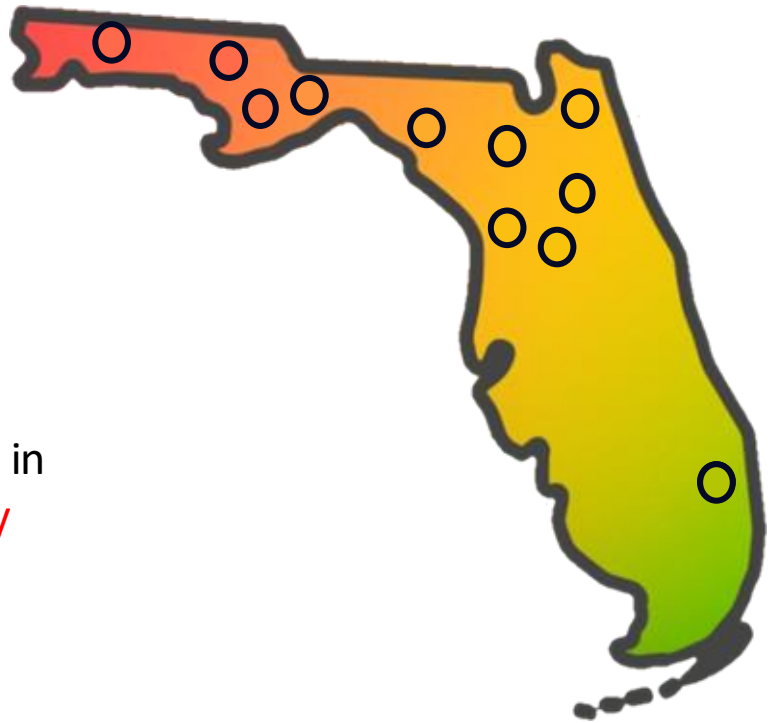
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CBI ≈ -1

Presences concentrated in low suitability habitats



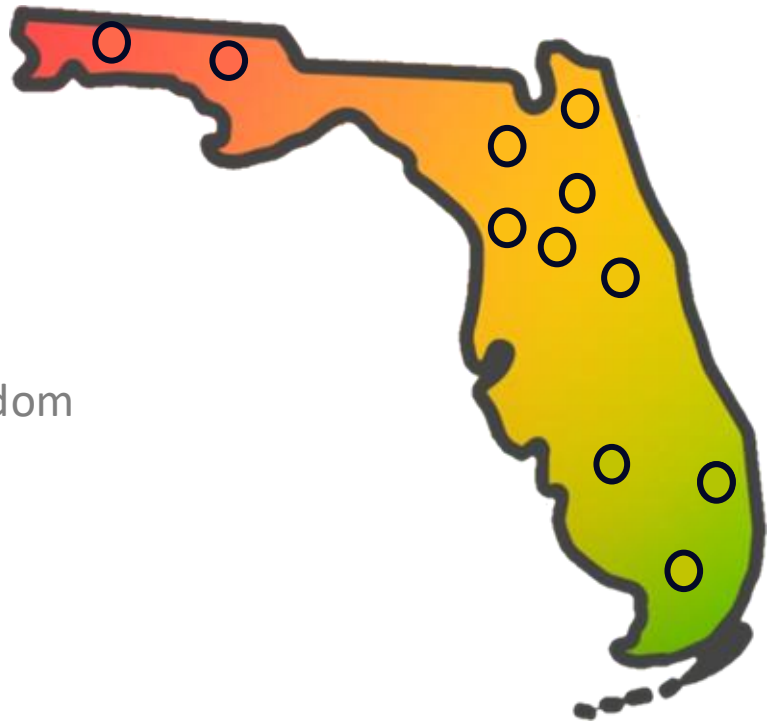
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$CBI \approx 0$

Suggests random performance

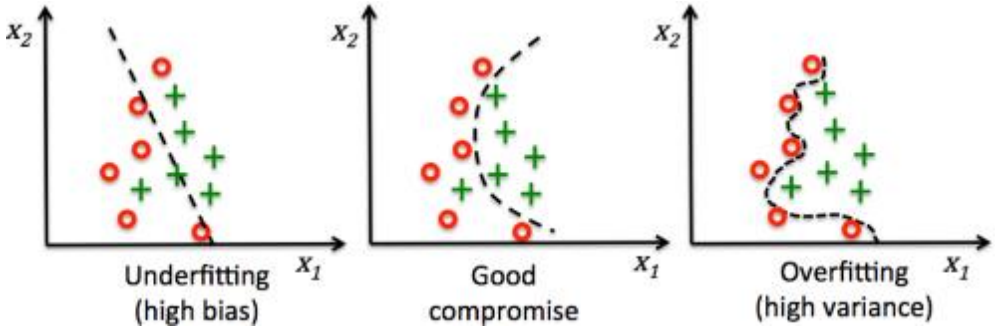


Feature classes and regularization multiplier

Feature type	Interpretation	Shape
Linear	Continuous variable	
Quadratic	Square of the variable	
Hinge	As threshold type, but response after the threshold (knot) is linear	

Balance complexity and fit

- Feature classes: shape of the expected response curve
- Response curve: the relationship between environmental variables and suitability
- Regularization multiplier: penalty applied to reduce overfitting



Sample code

Using tools in the ENMeval package

ENMevaluate generates models with different combinations of feature classes and RMs

```
eval <- ENMevaluate(
  occs = Galax_urceolata[, c("longitude", "latitude")],
  envs = vifStack,
  tune.args = list(fc = c("L", "Q"), rm = 1:2),
  partitions = "block",
  n.bg = 10000,
  parallel = FALSE,
  algorithm = 'maxent.jar',
)
```

Sample code

Using tools in the ENMeval package

Filter the resulting models by selecting for the one with the lowest omission rate and highest area under the curve value. Not ran but optional: select for highest CBI value

```
opt.seq <- results %>%
  filter(!is.na(AICc)) %>%                    # Exclude models with NA AICc
  filter(AICc == min(AICc)) %>%               # Minimum AICc
  filter(or.10p.avg != 0) %>%                 # Exclude zero omission
  filter(or.10p.avg == min(or.10p.avg)) %>%   # Minimum omission
  filter(auc.val.avg == max(auc.val.avg))     # Maximum AUC
```

filter(cbi.val.avg == max(cbi.val.avg))

How does ENMevaluate generate null models?

1 - Use the same number of presence points as the empirical model, but sample them randomly from the background

2 - Fit models with those random points using the same algorithm and environmental predictors

3 - Evaluate performance using real withheld presence data

4 - Repeat many times to generate a **null distribution** of performance metrics

5 - Compare the **empirical model's AUC, omission rate, and CBI** to the null distribution to assess significance

Sample code

Using tools in the ENMeval package

Generate null models using parameters from optimal model

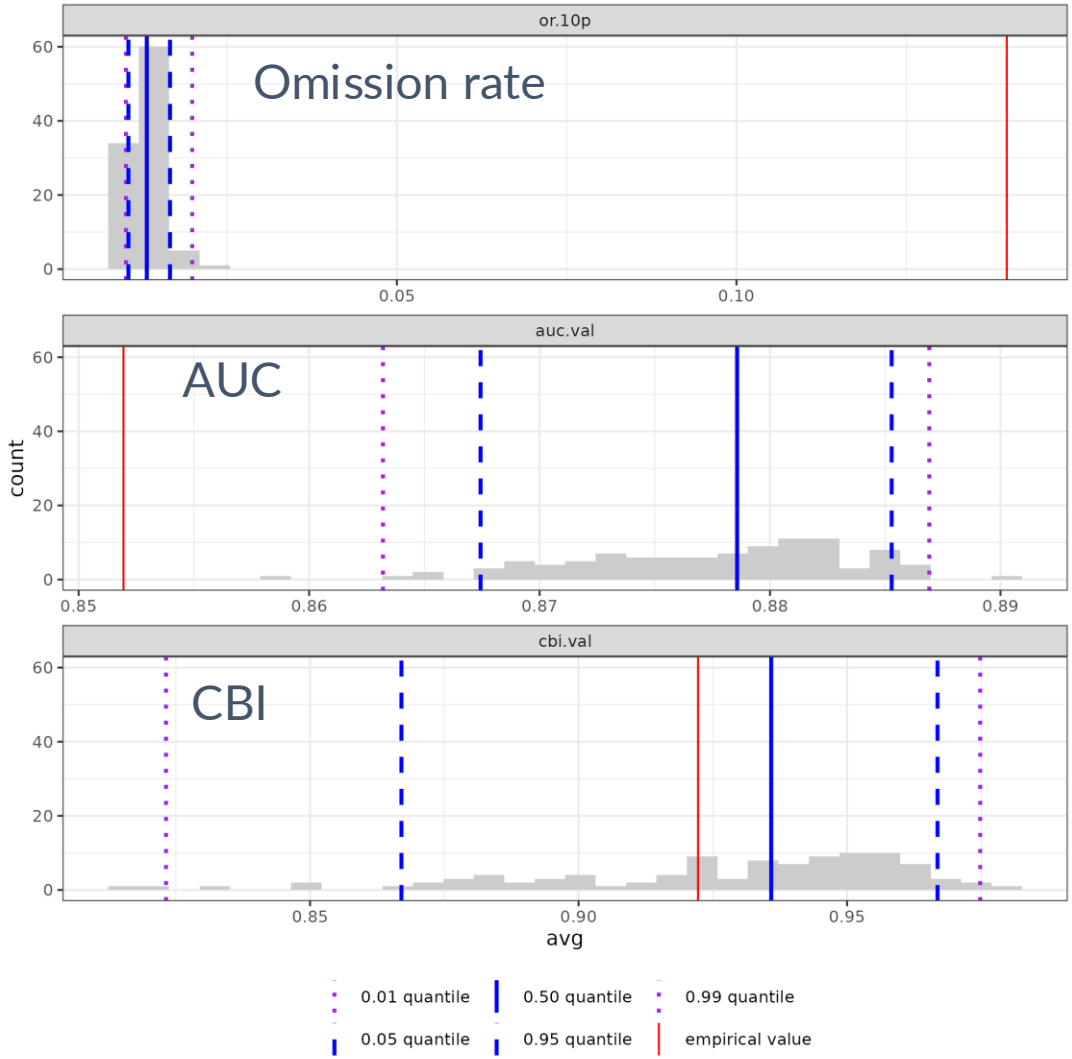
```
# Load optimal feature class (fc) and regularization multiplier (rm)
opt.seq <- read.delim("data/05_ENMs/Galax_urceolata_OptModel.txt")

# Extract parameters
fc <- opt.seq$fc
rm <- opt.seq$rm


# Run ENMnulls with optimal parameters and 100 iterations
spec.mod.null <- ENMnulls(
  eval,
  mod.settings = list(fc = fc, rm = rm),
  no.iter = 100
)
```

Output:

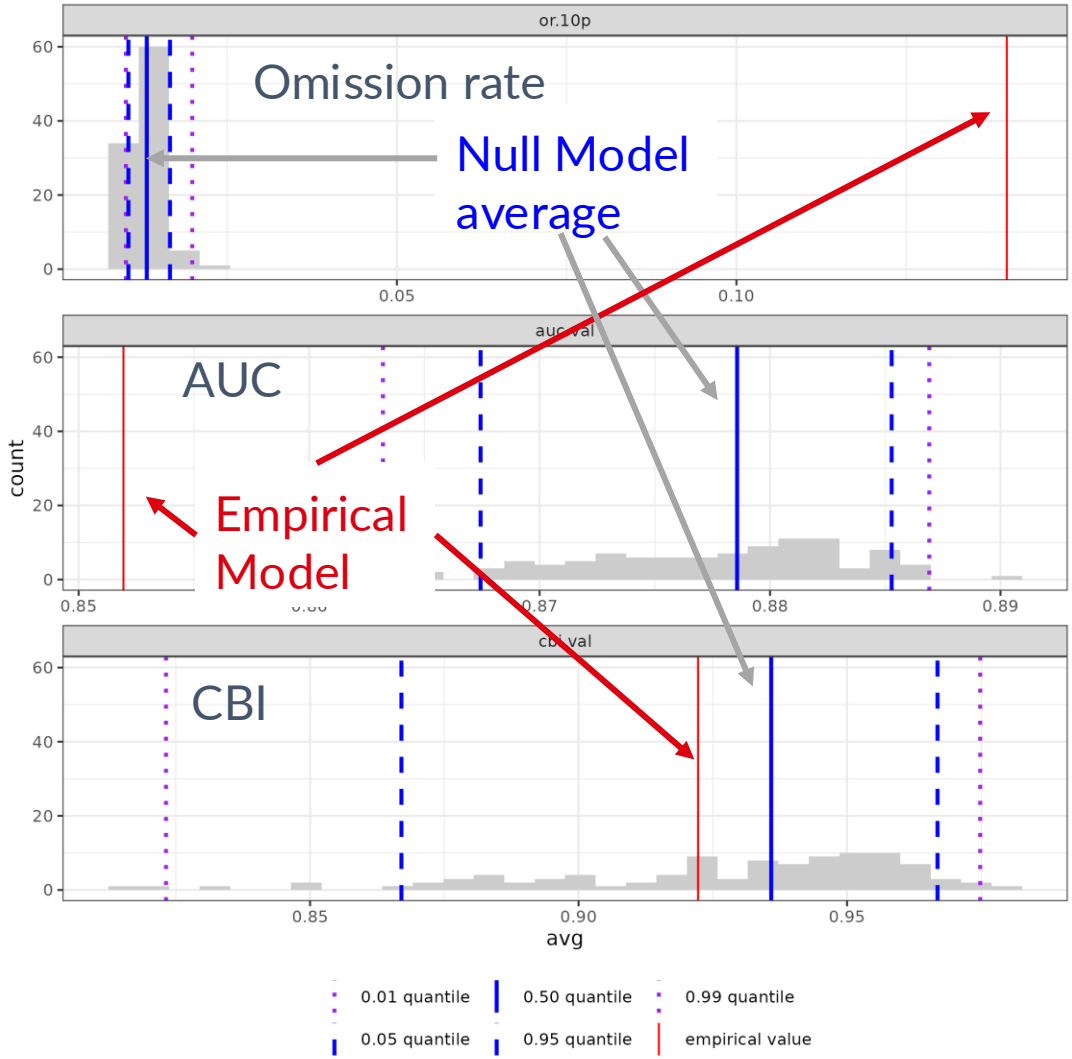
Plot shows the median AUC value, OR, and CBI for the empirical model (red) against the null distribution Quantiles (blue)



Output:

Plot shows the median AUC value, OR, and CBI for the empirical model (red) against the null distribution Quantiles (blue)

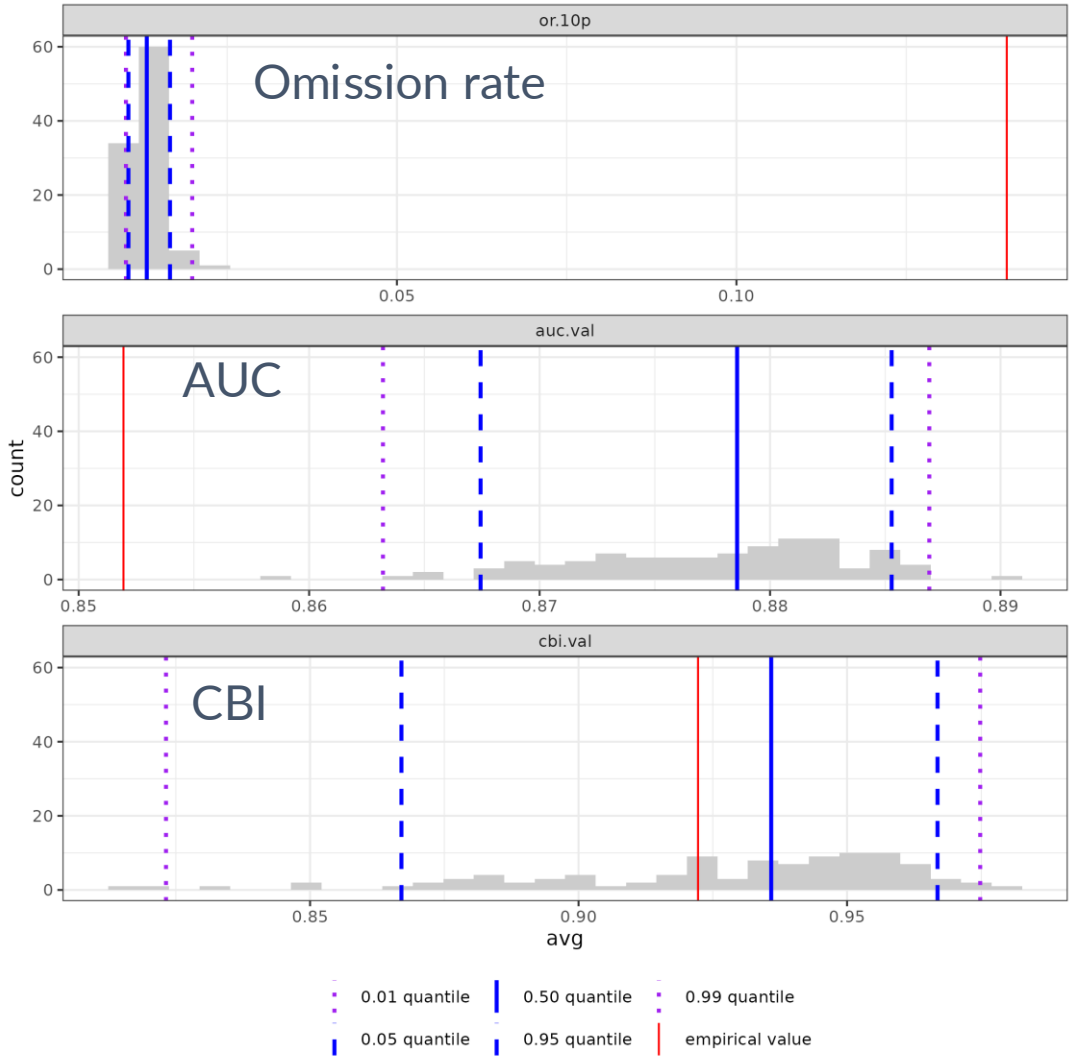
Meaning: If its **outside** the Null Distribution, then it differs **significantly from random**



Output:

Plot shows the median AUC value, OR, and CBI for the empirical model (red) against the null distribution Quantiles (blue)

Meaning: If its **outside** the Null Distribution, then it differs **significantly from random**



Sample code

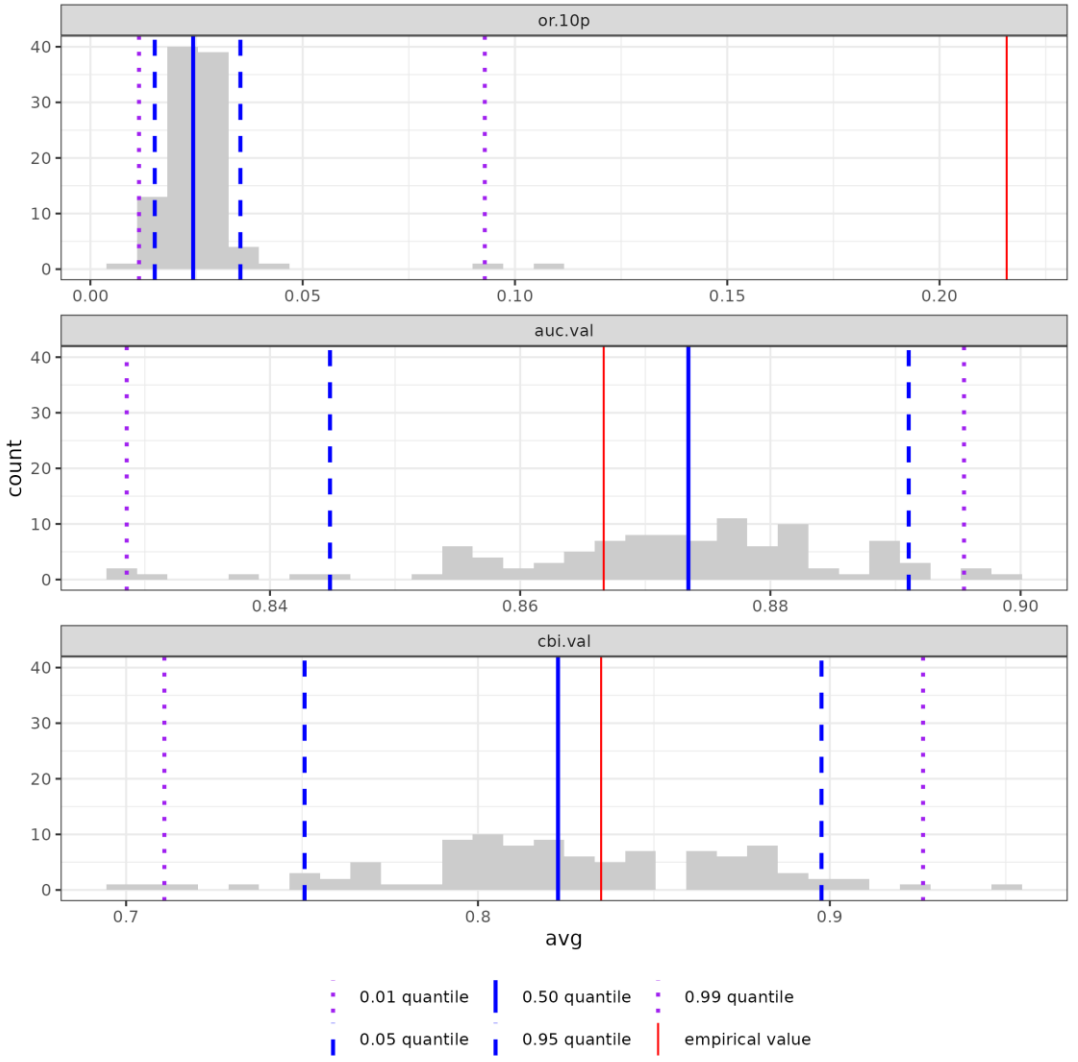
Using tools in the ENMeval package

Generate a new suite of models with a wider range of parameters

```
eval <- ENMevaluate(
  occs = Galax_urceolata[, c("longitude", "latitude")],
  envs = vifStack,
  tune.args = list(fc = c("L", "Q", "H", "LQ", "QH", "LH", "LQH"), rm = 1:5),
  partitions = "block",
  n.bg = 10000,
  parallel = FALSE,
  algorithm = 'maxent.jar',
)
```

Output:

New optimal model:
fc=LH, rm=1



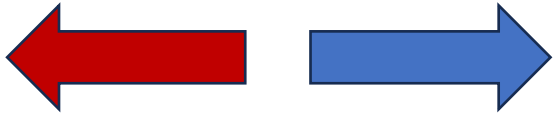
Why create null models?

- Determine whether model performance is better than expected by **chance**.
- Quantify how much better your model is than a null expectation.
- Avoid false confidence by evaluating for bias, spatial autocorrelation and overfitting
- Add rigor to model selection and interpretation

Some Caveats

- Null models rely on a balance between constraining randomizations
- Bohl et al. 2019 (what ENMnulls() methodology uses) uses fake occurrences from the background that are drawn w/out replacement.
- Fails to account for spatial autocorrelation and spatial biases, which may inflate Type 1 Error

Risk of Type 1 Error (Empirical has too easy of a time beating Null) Risks a bad model appearing to have strong ecological signal



Risk of Type 2 Error (Empirical has too difficult of a time beating Null) Risks a good model appearing to have weak ecological signal